Review of Model Categories

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Model categories are a key tool in the study of higher categories. By using various models we can handle the infinite amount of data that is always hidden inside a higher category This gives us an effective method to compare different models of higher categories as well as different models of presheaves. Famous introductions to model categories is the classic work by Hovey [Ho98], the very readable intro by Dwyer and Spalinski [DS95] and the more rigorous work by Hirschhorn [Hi03]

Definition of a Model Category

Let's start by reviewing the basic definitions regarding model categories. So, let's start with the definition of a model category and give some examples.

Definition 1.1. A *model category* is a category $\mathcal{M}$ with three distinguished classes of maps:

W *weak equivalences*

F *fibrations*

C *cofibrations*

each of which is closed under composition and contains all identity maps. A map that is simultaneously a weak equivalence and a fibration (resp. cofibration) will be called an *acyclic fibration* (resp. *acyclic cofibration*) or, alternatively, *trivial fibration* (resp. *trivial cofibration*). We demand that the following axioms are satisfied.

MC1 Small limits and small colimits exist in $\mathcal{M}$.

MC2 If f and g are maps in $\mathcal{M}$ such that gf is defined and if two of the three maps f, g, gf are weak equivalences, then so is the third.

MC3 If f is a retract of g and g is a fibration (resp. cofibration, weak equivalence), then so is f.

MC4 Given a diagram

$$\begin{tikzcd} A \arrow[r] \arrow[d, "i"'] & X \arrow[d, "p"] \\ B \arrow[r] & Y \end{tikzcd}$$

where either

(a) i is a cofibration and p is an acyclic fibration, or

(b) i is an acyclic cofibration and p is a fibration

a lift exists.

MC5 Any map f can be factored both as

(a) $f = pi$, where i is a cofibration and p is an acyclic fibration, and

(b) $f = p'i'$, where i is an acyclic cofibration and p is a fibration.

Fibrant and Cofibrant Objects. As an immediate consequence of MC1, a model category $\mathcal{M}$ has both an initial object $\varnothing$ and a terminal object $*$. An object Q of $\mathcal{M}$ is called *cofibrant* if the unique map $\varnothing \to Q$ is a cofibration. Similarly, and object R of $\mathcal{M}$ is called *fibrant* if the unique map $R \to *$ is a fibration.

The fibrant objects and cofibrant objects are ones of importance, as they are the ones that behave well with respect to certain homotopical constructions. Intuitively we think of cofibrant objects as the ones that can be easily mapped out of and of fibrant objects as objects which we can easily map into. This can clearly be seen by looking at the lifting diagram in MC4. We will thus want a way to approximate arbitrary objects by fibrant/cofibrant ones.

Definition 1.2. Let X be an object in a model category $\mathcal{M}$. A *cofibrant approximation* of X is an object $\widetilde{X}$ together with a weak equivalence $i : \widetilde{X} \to X$. Similarly, a *fibrant approximation* of X is an object $\widehat{X}$ together with a weak equivalence $p : X \to \widehat{X}$.

Note that by applying MC5 to the maps $\varnothing \to X$ and $X \to *$, we have that cofibrant approximations and fibrant approximations always exist. It is a fact that fibrant (resp. cofibrant) approximations of an object are unique up to a weak equivalence.

Definition 1.3. Let $f : X \to Y$ be a morphism in a model category $\mathcal{M}$. A *cofibrant approximation* of $f : X \to Y$ is a diagram

$$\begin{tikzcd} \widetilde{X} \arrow[r, "\widetilde{f}"] \arrow[d, "i"'] & \widetilde{Y} \arrow[d, "j"] \\ X \arrow[r, "f"'] & Y \end{tikzcd}$$

in which $i : \widetilde{X} \to X$ and $j : \widetilde{Y} \to Y$ are cofibrant approximations. Similarly, a *fibrant approximation* of $f : X \to Y$ is a diagram

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "p"'] & Y \arrow[d, "q"] \\ \widehat{X} \arrow[r, "\widehat{f}"'] & \widehat{Y} \end{tikzcd}$$

in which $p : X \to \widehat{X}$ and $q : Y \to \widehat{Y}$ are fibrant approximations.

Again, by applying MC5, one can show that cofibrant approximations and fibrant approximations of a morphism always exist.

Examples The quintessential model category is **Top**, the category of topological spaces and continuous maps, with the Quillen model structure. A map $f : X \to Y$ is a

W weak equivalence if it is a weak homotopy equivalence.

F fibration if it is a Serre fibration.

C cofibration if it is a retract of a relative CW-complex.

Every object in this model category is fibrant. Cofibrant objects include CW-complexes. We can take cofibrant approximations by cellular approximation.

Another example of a model category is the category $\mathbf{Ch}_R^+$ of nonnegatively graded chain complexes of R-modules. In this model structure, a chain map $f_\bullet : X_\bullet \to Y_\bullet$ is a

W weak equivalence if it is a quasiisomorphism.

F fibration if, for $k \geq 1$, $f_k : X_k \to Y_k$ is an epimorphism.

C cofibration if, for $k \geq 0$, $f_k : X_k \to Y_k$ is a monomorphism with projective cokernel.

Every object in the model category is fibrant. Cofibrant objects are chain complexes of projective modules. This example is of interest because the homotopy theory chain complexes turns out to be homological algebra. This indicates that model categories give us a way of doing "homotopical algebra".

Properness There are several important properties that a model category can have. One of them is being simplicial, which will be discussed in section 4. Another one is the notion of properness.

Definition 1.4. A model category is called left proper, if in the following push out diagram:

$$\begin{tikzcd} A \arrow[r, "i"] \arrow[d, "j"'] & B \arrow[d, "k"] \\ C \arrow[r, "l"] & D \end{tikzcd}$$

i being a cofibration and j being an equivalence implies that k is also a weak equivalence. Essentially we say, push out along a cofibration preserves weak equivalences.

Note that this is always true if the objects are cofibrant. In particular, any model category in which all objects are cofibrant is left proper.

Definition 1.5. Similarly we say a model category is right proper, if in the following pull back diagram:

$$\begin{tikzcd} A \arrow[r, "j"] \arrow[d, "q"'] & B \arrow[d, "p"] \\ C \arrow[r, "i"] & D \end{tikzcd}$$

p being a fibration and i being an equivalence implies that j is also a weak equivalence. Essentially we say, pullback along a fibration preserves weak equivalences.

Note that this is always true if the objects are fibrant. In particular, any model category in which all objects are fibrant is right proper.

Definition 1.6. A model category is called proper if it is right proper and left proper.

Example 1.1. In the model category **Top** every object is fibrant, so it is definitely right proper. It turns out it is also left proper, which means **Top** is actually proper.

Example 1.2. Similar to this the model structure on $\mathbf{Ch}_R^+$ is also proper.

As we move along we will see examples of model structures that are not proper.

Cartesian Model Categories. There is one last concept that will show up as we move along and that is Cartesian model category.

Definition 1.7. Let $\mathcal{M}$ be a model category such that the underlying category is cartesian closed. Let $i : A \to B$ and $j : C \to D$ be cofibrations and $p : X \to Y$ a fibration. Then we say it is *Cartesian* if it satisfies one of following equivalent conditions:

- The map $A \times D \coprod_{A \times C} B \times C \to B \times D$ is a cofibration which is a acyclic cofibration if either i or j are.

- The map $X^B \to Y^B \times_{Y^A} X^A$ is a fibration, which is a acyclic fibration if either i or p are acyclic.

Quillen Adjunctions and Quillen Equivalences

Quillen adjunctions are adjunctions between model categories that play well with the model structure. The big idea is that a Quillen adjunction presents an adjunction between the presented ∞-categories.

Definition 2.1. Let $\mathcal{M}$ and $\mathcal{N}$ be model categories. An adjunction $F : \mathcal{M} \rightleftarrows \mathcal{N} : G$ is called a *Quillen adjunction* if any of the following equivalent conditions holds:

1. F preserves cofibrations and G preserves fibrations.
2. F preserves cofibrations and acyclic cofibrations.
3. G preserves fibrations and acyclic fibrations.

An example of a Quillen adjunction is given by

$$- \otimes A : \mathbf{Ch}_{\mathbb{Z}}^{+} \rightleftarrows \mathbf{Ch}_{\mathbb{Z}}^{+} : \mathrm{Hom}_{\mathbb{Z}}(A, -)$$

for some abelian group A.

Definition 2.2. Let $F : \mathcal{M} \rightleftarrows \mathcal{N} : G$ be a Quillen adjunction. The adjunction is called a *Quillen equivalence* if, for every cofibrant object A of $\mathcal{M}$ and every fibrant object X of $\mathcal{N}$, a map $F(A) \to X$ is a weak equivalence if and only if the adjoint map $A \to G(X)$ is a weak equivalence.

Equivalently, we have the following characterization.

Theorem 2.1. *A Quillen adjunction $F : \mathcal{M} \rightleftarrows \mathcal{N} : G$ is a Quillen equivalence if and only if the following two conditions are satisfied:*

1. *For every object cofibrant object $A \in \mathcal{M}$, the derived unit map $A \to GF(A) \to GR(F(A))$ is a weak equivalence in $\mathcal{M}$. Here $R(F(A))$ is the fibrant replacement of $F(A)$ inside $\mathcal{N}$.*
2. *For every object fibrant object $B \in \mathcal{N}$, the derived unit map $FQ(G(B)) \to FG(B) \to B$ is a weak equivalence in $\mathcal{N}$. Here $Q(G(B))$ is the cofibrant replacement of $G(B)$ inside $\mathcal{M}$.*

Remark 2.1. Note that many properties are not preserved by Quillen equivalences. For example, a right proper model category can be Quillen equivalent to a left proper one.

We will give an example of a Quillen equivalence below.

Simplicial Sets and the Kan Model Structure

Definition 3.1. Let $\mathbf{\Delta}$ be the category with objects $[\mathbf{n}] = \{0, \ldots, n\}$, $n \geq 0$, and morphisms nondecreasing functions. A *simplicial set* is a functor $\mathbf{\Delta}^{\mathrm{op}} \to \mathbf{Set}$. A morphism between simplicial sets is simply a natural transformation. The category of simplicial sets and morphisms will be denoted **sSet**.

We have a functor $S_\bullet : \mathbf{Top} \to \mathbf{sSet}$ taking a topological space X to the simplicial set $S_\bullet X$, where

$$S_n X = \{f : \Delta^n \to X\}.$$

On the other hand, we have a functor $|-| : \mathbf{sSet} \to \mathbf{Top}$, taking a simplicial set $Y_\bullet$ to its geometric realization

$$|Y_\bullet| = \int^{[\mathbf{n}] \in \mathbf{\Delta}} Y_n \times \Delta^n.$$

These functors give us an adjunction $|-| : \mathbf{sSet} \rightleftarrows \mathbf{Top} : S_\bullet$. This adjunction actually gives us a way to transport the model structure from **Top** to one on **sSet**, called the Kan model structure.

Definition 3.2. There is a model structure on **sSet**, called the Kan model structure. A map $f_\bullet : X_\bullet \to Y_\bullet$ is a

W weak equivalence if $|f_\bullet| : |X_\bullet| \to |Y_\bullet|$ is a weak equivalence of topological spaces.

F fibration if it has the Kan extension property.

C cofibration if it is an inclusion.

Every object in this model category is cofibrant. The fibrant objects are exactly the Kan complexes.

Remark 3.1. If we give **sSet** the Kan model structure and **Top** the Quillen model structure model structure the the adjunction

$$|-| : \mathbf{sSet} \rightleftarrows \mathbf{Top} : S_\bullet$$

becomes a Quillen equivalence.

The Kan model structure is probably the most user-friendly model structure in the world and a lot of effort in homotopy theory is to reduce complicated questions to the case where it can be studied in the Kan model structure. In particular, it is proper, cartesian and simplicial (which we will discuss in the next section).

A very detailed discussion of simplicial sets and the Kan model structure can be found in [GJ99] which dedicates a whole book to this topic.

Simplicial Model Categories

Definition 4.1. A *simplicial category* is a category $\mathcal{C}$ together with

- a simplicial set $\mathrm{Map}(X, Y)$ for every pair of objects $X, Y \in \mathcal{C}$.
- a composition rule $c_{X,Y,Z} : \mathrm{Map}(Y, Z) \times \mathrm{Map}(X, Y) \to \mathrm{Map}(X, Z)$ for every triple of objects $X, Y, Z \in \mathcal{C}$.
- a morphism $i_X : * \to \mathrm{Map}(X, X)$ for every object $X \in \mathcal{C}$.
- an isomorphism $\mathrm{Map}(X, Y)_0 \cong \mathcal{C}(X, Y)$ that commutes with the composition rule.

We demand that certain coherence axioms are satisfied. (Associativity, left unit, right unit.)

A basic example of a simplicial category is **sSet** itself.

Definition 4.2. A *simplicial model category* is a category $\mathcal{M}$ that is simultaneously a simplicial category and a model category such that

- For every simplicial set K and every object X of $\mathcal{M}$, we have objects $X \otimes K$ and X^K of $\mathcal{M}$ such that there are isomorphisms

  $$\mathrm{Map}(K, \mathrm{Map}(X, Y)) \cong \mathrm{Map}(X \otimes K, Y) \cong \mathrm{Map}(X, Y^K)$$

  that are natural in X, YK.
- for every fibration $p : X \to Y$ and every cofibration $i : A \to B$, The natural map

  $$\mathrm{Map}(B, X) \to \mathrm{Map}(A, X) \times_{\mathrm{Map}(A,Y)} \mathrm{Map}(B, Y)$$

  is a fibration, and a weak equivalence if either $i : A \to B$ or $p : X \to Y$ are.

We can take **sSet** with the Kan model structure as an example of a simplicial model category.

More details on this can be found in chapter 9 of Hirschhorn [Hi03]

Simplicial Spaces and the Reedy Model Structure

Definition 5.1. A *simplicial space* is a simplicial object of **sSet**. That is, a simplicial space is a functor $\mathbf{\Delta}^{\mathrm{op}} \to \mathbf{sSet}$. A morphism of simplicial spaces is a natural transformation. The category of simplicial spaces and morphisms will be denoted **sSpace**.

We can identify the category **sSet** as full subcategory of **sSpace** consisting of constant functors $\mathbf{\Delta}^{\mathrm{op}} \to \mathbf{sSet}$. In particular, for any simplicial space X and any simplicial set K, we can form the simplicial space $X \times K$. This lets us define a simplicial set $\mathrm{Map}(X, Y)$ where

$$\mathrm{Map}(X, Y)_n = \{f : X \times \Delta[n] \to Y\}.$$

This construction, together with appropriate composition and identity rules, gives **sSpace** the structure of a simplicial category.

We can equip **sSpace** with a model structure, called the Reedy model structure. A map $f : X \to Y$ in the category **sSpace** is a

- weak equivalence if it is a degree-wise weak equivalence.
- fibration (resp. acylic fibration) if, for every $k \geq 0$, the natural map

  $$\mathrm{Map}(F(k), X) \to \mathrm{Map}(\dot{F}(k), X) \times_{\mathrm{Map}(\dot{F}(k),X)} \mathrm{Map}(F(k), Y)$$

  is a fibration (resp. acyclic fibration), where $F(k)$ is the simplicial space $\mathrm{Map}(-, \Delta[k])$, $\dot{F}(k)$ is the largest subobject of $F(k)$ that does not contain $\mathrm{id}_{\Delta k} \in F(k)_k = \mathrm{Map}(\Delta[k], \Delta[k])$.
- cofibration if it is an inclusion.

This model category has nice properties, such as compact generation (which gives us functorial cofibrant and fibrant replacements) and properness. By taking $X \otimes K = X \times K$ and $X^K = \mathrm{Map}(K, X)$, we have that this gives **sSpace** a simplicial model category structure. Moreover, like the Kan model structure, it is also a Cartesian model category.

The Reedy model structure was introduced in far greater generality in [ReXX]. There is also a nice discussion of it in [DGS93] (section 2).

Bousfield Localization

One of the key constructions in the world of model categories are *Bousfield localizations*. It allows us to modify a given model category in a way that a desired map becomes an equivalence. Bousfield localizations can be applied to a broad range of model categories, but our focus is on simplicial sets and simplicial spaces. Thus we will only discuss it in that context. We will state it in the context of simplicial spaces and the simplicial set case is completely analogeous. For a more general approach the most rigorous source is certainly the detailed work by Hirschhorn ([Hi03]).

Theorem 6.1. *Let* **sSpace** *be the category of simplicial spaces with the Reedy model structure. Moreover, let $f : A \to B$ be a map in* **sSpaces**. *There is a model structure on* **sSpace** *called the localization model structure, which satisfies the following conditions:*

- *An object W is fibrant if it is Reedy fibrant and if the simplicial map*

$$Map(B, W) \to Map(A, W)$$

  is a Kan equivalence.

- *A map $g : C \to D$ is a weak equivalence if for every fibrant object W, the map*

$$Map(D, W) \to Map(C, W)$$

  is a Kan equivalence.

- *Cofibrations are just inclusions*

- *Fibrations are maps which satisfy the lifting conditions with repect to acyclic cofibrations.*

- *Weak equivalences (fibrations) between fibrant objects are just Reedy weak equivalences (fibrations).*

Using Bousfield localizations we will be able to start with some fairly simple model structures (like the Reedy model structure) and derive some very important and complicated model structures (such as complete Segal space model structure, which models higher categories).

References

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